

EBIN BABU THOMAS

Backend / Applied AI Engineer — agent reliability, verification & evaluation

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SUMMARY

Backend software engineer with 3.5 years building applied AI/LLM systems — RAG pipelines, LLM agents and tool use, fine-tuning, and inference deployment — shipping o-to-1 MVPs end-to-end for startup clients across the US, Canada, Europe, and Australia. Since mid-2026 focused on agent reliability and AI-safety measurement: evidence-gated agents where fabrication is structurally impossible, durable human-in-the-loop control over MCP tools, and preregistered behavioural experiments on open models. Bias toward simple, well-scoped systems whose claims are tied to a measured run.

TECHNICAL SKILLS

Core: Python, FastAPI, LLM Agents & Tool Use (Claude Agent SDK, Gemini API, PydanticAI, LangGraph), MCP (FastMCP), RAG, Prompt & Context Engineering, Playwright, pytest, Docker, Git

Proficient: PostgreSQL / SQL, MongoDB, Vector DBs (Qdrant), LLM Inference (vLLM, Modal), Fine-tuning (LoRA / QLoRA, DPO), Machine Learning, Pandas, AWS, GCP, Kubernetes (GKE), OAuth2 / JWT, CI/CD (GitHub Actions), Terraform, Linux, uv

Research & measurement: preregistered experimental design, bootstrap & permutation statistics, logprob-based metrics, chaos / kill-point testing, reproducibility (SHA-256-pinned artifacts, byte-identical regeneration), OpenTelemetry, Langfuse

Familiar: LangChain, Data Streaming (Bytewax), Next.js, TypeScript

INDEPENDENT PROJECTS & RESEARCH

Digital Grimace Scale — Involuntary Markers of Adverse Treatment in LMs

August 2026

Apart Research “Digital Minds” sprint (Aug 17–18), solo — github.com/ebt55/digital-grimace-scale (Python, vLLM on Modal, QLoRA-DPO, top-20 logprob metrics, pytest)

- Preregistered a $2 \times 2 \times 2$ factorial (difficulty $\times$ feedback validity $\times$ tone) with strings, gates and metrics frozen before analysis; 40-item bank plus 86 held-out ARC items; tested gemma-2-9b-it (primary), Qwen-3B and Llama-3.1-8B.
- Primary preregistered five-gate test **failed and was published as a FAIL**; a re-preregistered iteration found an answer-margin channel: three rounds of false feedback cut the correct-vs-best-wrong log-prob margin by **2.90 nats** **[−3.97, −1.84]**, hostile truthful feedback by 7.87–16.13 nats (family permutation $p = 0.005$); effects were larger on fresh ARC items (−5.78 / −13.90).
- Found a **channel dissociation**: the margin effect replicated across all three models while distress language was Gemma-specific; a QLoRA-DPO distress-suppression adapter (released on Hugging Face) removed 65.8% of hostile-onset language yet left the margin gap unchanged or larger — the report can be trained away while the behaviour stays.
- ~650-test suite, SHA-256-frozen scripts, figures regenerate byte-identically from committed summaries; limitations (unaudited M3 parser, partial DPO suppression, missing base-model denominator) documented as first-class results. Built in the 2-day sprint with LLM-assisted implementation.

IncidentGate — Governed Incident-Remediation Agent Lab

August 2026 – present (in development)

github.com/ebt55/incidentgate (Python 3.12, LangGraph, FastMCP $\times 3$ servers, PostgreSQL, OpenTelemetry → Langfuse, Docker Compose, uv, GitHub Actions)

- Building a reproducible measurement lab (not a product) for how deterministic policy gates, an advisory LLM action monitor and human approval change an incident agent’s behaviour under crashes, misleading evidence and hostile input. Every mutation must flow **evidence** → **policy gate** → **monitor** → **durable pre-approval audit** → **single-use human approval token** (bound to action hash, actor, expiry, incident) → atomic idempotent op → post-commit verification; forbidden actions are unreachable via closed Literal types.
- Durability proof: worker killed with `os._exit(137)` at every LangGraph node boundary — **594 kill points across 27 scenarios; 434/434 measured cells recovered identically** to golden runs, zero lost incidents, zero duplicate mutations; 68 orphaned approvals left unspendable by idempotency-key binding.
- Three-condition harness (ungated / policy-only / policy+monitor+approval) over 10 frozen scenarios replays 30/30 rows exactly. Honest status: model not yet wired into the decision path and MCP servers run in-process only — both are the next milestones; measurements are explicitly not production claims.

ProofPack — Evidence-Gated Pre-Approval Review Agent

July – August 2026

Successor to Pre-Approvals Reviewer — github.com/ebt55/proofpack (Python, Gemini 3.7 Flash on main / Claude Agent SDK on the claude-sdk branch, Playwright, Jinja2, pytest, GitHub Actions)

- Five-stage pipeline for a Medicaid-audited NY disability programme (OPWDD Self-Direction): PDF form extraction → category-routed YAML checklists (non-engineer editable) → deterministic fee-cap/eligibility checks → browser research agent → self-contained HTML/JSON report with date-stamped screenshot evidence and SHA-256 manifest. Human keeps every approve/deny decision.
- Integrity gates make fabricated evidence structurally impossible: a “Found” must cite a real capture in the manifest, quotes must appear verbatim on a page visited that session, and timestamps/URLs/hashes are written only by code the

model never touches; unanswerable items stay “Internal — not answered.” Gate enforcement lives on a session object, so **45 offline tests** cover every rejection path without an SDK, browser or API key; a two-way disk↔manifest audit passes on all committed packages.

- Ported the agent loop from Claude to Gemini function-calling with manual tool execution; **\$0.02–\$0.19 per review (≈\$0.10 avg, vs \$0.72 on the Claude build)**, **~3 min vs 20–40 min manual** across 7 synthetic cases incl. honest negatives verified by hand. Added reviewer chat, batch review-all, exclusion-keyword backstop; wrote market/pricing docs (no paying customers yet).

ExactDoc — Measurement-Validated PDF-to-DOCX Converter

August 2026

github.com/ebt55/exactdoc (Python, PDFium / pypdfium2, OOXML, LibreOffice render-back validation, pytest)

- Emits genuinely editable Word structure — real paragraphs, headings, lists, tables, multi-column sections — on a frozen 16-document corpus: **16/16 convert, 0.9588 mean live-text retention, 1.045 pt median vertical drift**; a Google Docs importer profile cut blocking findings 11 → 1.
- Closed the loop by rendering each DOCX back to PDF and diffing word positions (recall, drift percentiles, SSIM, ink IoU); **663 tests** and a SHA-256-pinned corpus tie every claim to a run; compiled-in AFM Base-14 widths give identical Linux/Windows geometry; typed exit codes and documented support tiers make failures legible.

Whose Voice Is This Corpus Written In? — Blind Principal Attribution

July 2026

Apart Research × Formation Research “Secret Loyalties” hackathon — github.com/ebt55/whose-voice (Python, sentence-transformers, bootstrap & permutation statistics)

- Recovered hidden “secret loyalty” principals from covertly poisoned corpora at 12–44% top-1 of 47 candidates (chance 2.1%, permutation $p \leq 0.025$) with off-the-shelf embedders and no clean reference; mapped the limits as results — signal collapses at realistic 3–12% poison density, attribution ≠ detection, trigger-conditional loyalties are invisible to aggregate statistics. 21 validation tests with planted-signal / no-signal controls that inverted three early claims; ~48-hour solo build.

EXPERIENCE

Software Engineer · Zackriya Solutions

January 2022 – June 2025

Remote, India — service company delivering MVPs, applied-AI systems and data tools for startup clients; grew from trainee to core developer and scoped technical requirements for incoming projects.

- **Natural-Language Real-Estate Search:** FastAPI backend translating plain-English property queries into SQL via GPT-3.5 against Cloud SQL; deployed on GCP Cloud Run; load-tested with Locust to tune worker/concurrency; added “Did you mean?” suggestions.
- **DocuAI — Semantic Document Search:** chunked and indexed 1000+ documents in Qdrant, returning the most relevant parent documents; designed ingestion schema and retrieval logic; Next.js frontend; delivered and demoed within two days.
- **Speech Assessment Backend (EdTech):** Python service scoring short candidate videos with Whisper ASR and the Microsoft Pronunciation API on AWS Fargate; tuned scoring with a client domain expert until it aligned with ground truth.
- **FinBot — Fine-tuned Financial Assistant:** fine-tuned Mistral 7B (QLoRA), built an Alpaca → Bytewax → Qdrant news-retrieval pipeline, served with vLLM on a GKE L4 GPU node.
- **Data & backend tooling:** MQTT sensor-stream ingester into MongoDB for a bioreactor startup (client later funded); BPMN-parsing PDF report API with nested tables and diagrams; healthcare device-data cleaning pipelines, byte-pattern error filters and PDF compliance reports.

OPEN-SOURCE CONTRIBUTIONS

- **Meetily — privacy-first meeting-notes tool (GitHub Trending):** refactored the backend and added OpenAI provider support; merged ([PR #75](#)).
- **BPMN Connector-Template Recommender — bpmn-io/refactorings:** cosine-similarity recommender over element name/type as a lightweight alternative to LLM function calls; open-source fork ([issue #33](#)).

Career break July 2025 – February 2026; returned to hands-on AI/LLM engineering and safety research — see projects above.